

# Winter Weather Panel: Descriptive Report

PRISM county-month data, 1981–2025

2026-08-28

**Provenance.** All exhibits generated by `codePYTHON/09_descriptive_weather_full.ipynb`, executed on the Kodama server against the complete PRISM county-month panel. Figures are vector PDFs. Tier 1 QA exhibits are indexed in Appendix B rather than included/reproduced here in the report.

**Status.** The warm-anomaly exhibits in Section 5 are **provisional** pending confirmation of the anomaly definition. Section 8 lists the choices requiring a decision.

# Contents

|          |                                              |           |
|----------|----------------------------------------------|-----------|
| <b>1</b> | <b>Coverage and data integrity</b>           | <b>3</b>  |
| <b>2</b> | <b>Long-run winter warming</b>               | <b>5</b>  |
| <b>3</b> | <b>Where winters have warmed</b>             | <b>6</b>  |
| <b>4</b> | <b>Cold extremes</b>                         | <b>8</b>  |
| <b>5</b> | <b>Anomalously warm winters</b>              | <b>9</b>  |
| <b>6</b> | <b>Distributional shift</b>                  | <b>12</b> |
| <b>7</b> | <b>Robustness: PRISM versus ERA5</b>         | <b>14</b> |
| <b>8</b> | <b>Decisions requested</b>                   | <b>15</b> |
| <b>A</b> | <b>State-level winter-temperature trends</b> | <b>16</b> |
| <b>B</b> | <b>Tier 1 QA artefacts (not reproduced)</b>  | <b>23</b> |

# 1 Coverage and data integrity

The panel is complete: no duplicate keys, no unmatched rows across the monthly and derived files, no incomplete county-months, and no QA-field mismatches. The excluded county-winters are exactly the two boundary winters that cannot be complete, not unexplained missingness.

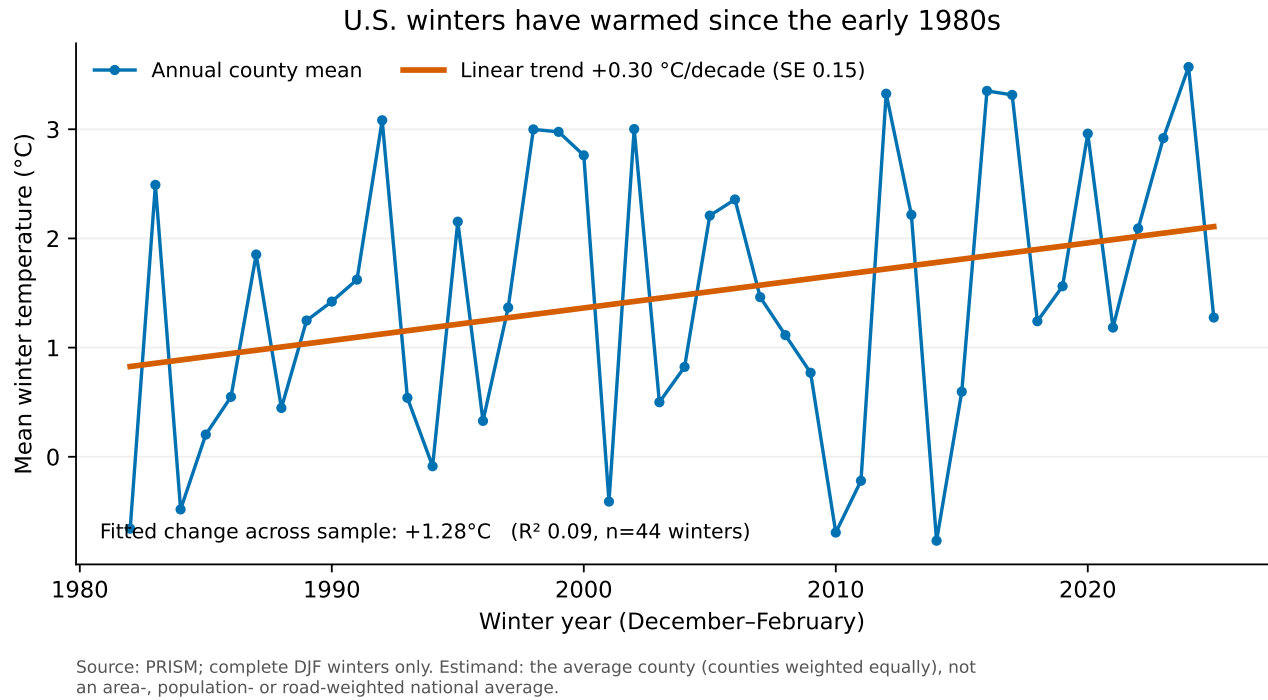
**Table 1:** Coverage and integrity diagnostics

| Metric                             | Value     |
|------------------------------------|-----------|
| dataset                            | PRISM     |
| monthly rows                       | 1678320   |
| derived rows                       | 1678320   |
| merged rows                        | 1678320   |
| counties                           | 3108      |
| first year                         | 1981      |
| last year                          | 2025      |
| duplicate monthly keys             | 0         |
| duplicate derived keys             | 0         |
| monthly only rows                  | 0         |
| derived only rows                  | 0         |
| incomplete months                  | 0         |
| qa mismatch n days                 | 0         |
| qa mismatch expected days          | 0         |
| qa mismatch is incomplete          | 0         |
| qa mismatch dataset types          | 0         |
| candidate county winters           | 142968    |
| complete county winters            | 136752    |
| excluded incomplete county winters | 6216      |
| first complete winter              | 1982      |
| last complete winter               | 2025      |
| expected calendar years            | 1981–2025 |
| observed calendar years            | 1981–2025 |
| observed complete winters          | 1982–2025 |

**Table 2:** Tier 1 QA findings

| Check                                       | Result | Status |
|---------------------------------------------|--------|--------|
| Duplicate county-year-month keys            | 0      | PASS   |
| Rows appearing in only one merge source     | 0      | PASS   |
| Incomplete county-months                    | 0      | PASS   |
| Excluded incomplete county-winters          | 6216   | INFO   |
| County-variable mean flags ( $ z  \geq 5$ ) | 20     | REVIEW |

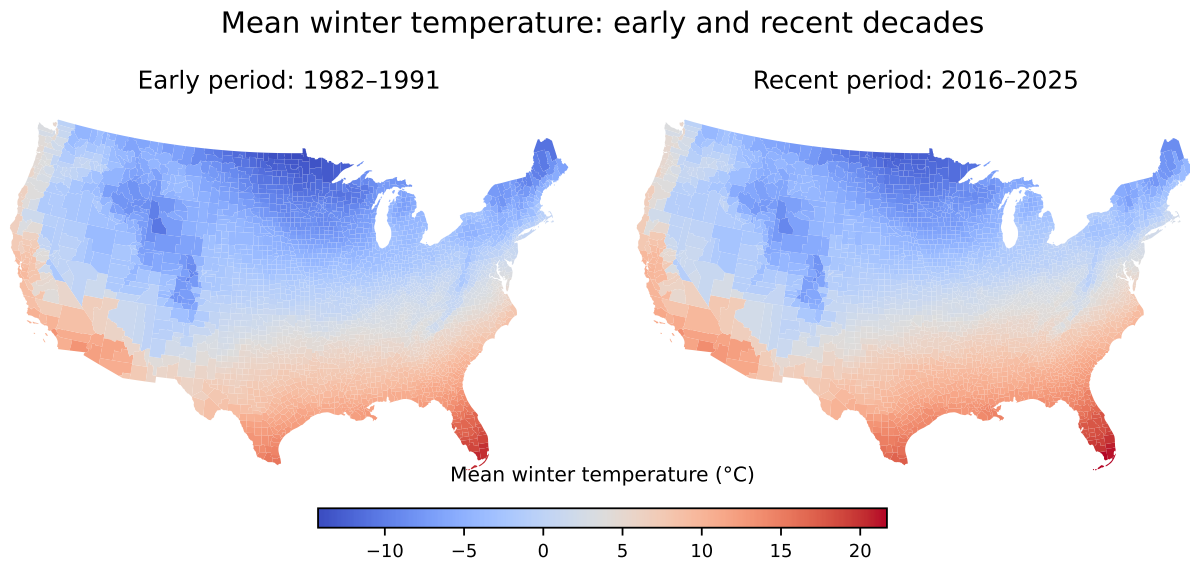
## 2 Long-run winter warming



**Figure 1:** National winter temperature, 1982–2025. Each point is the unweighted mean across counties, so the estimand is the average county rather than an area- or population-weighted national average. The fitted slope is shown with its standard error.

State-level trends are reproduced in Appendix A.

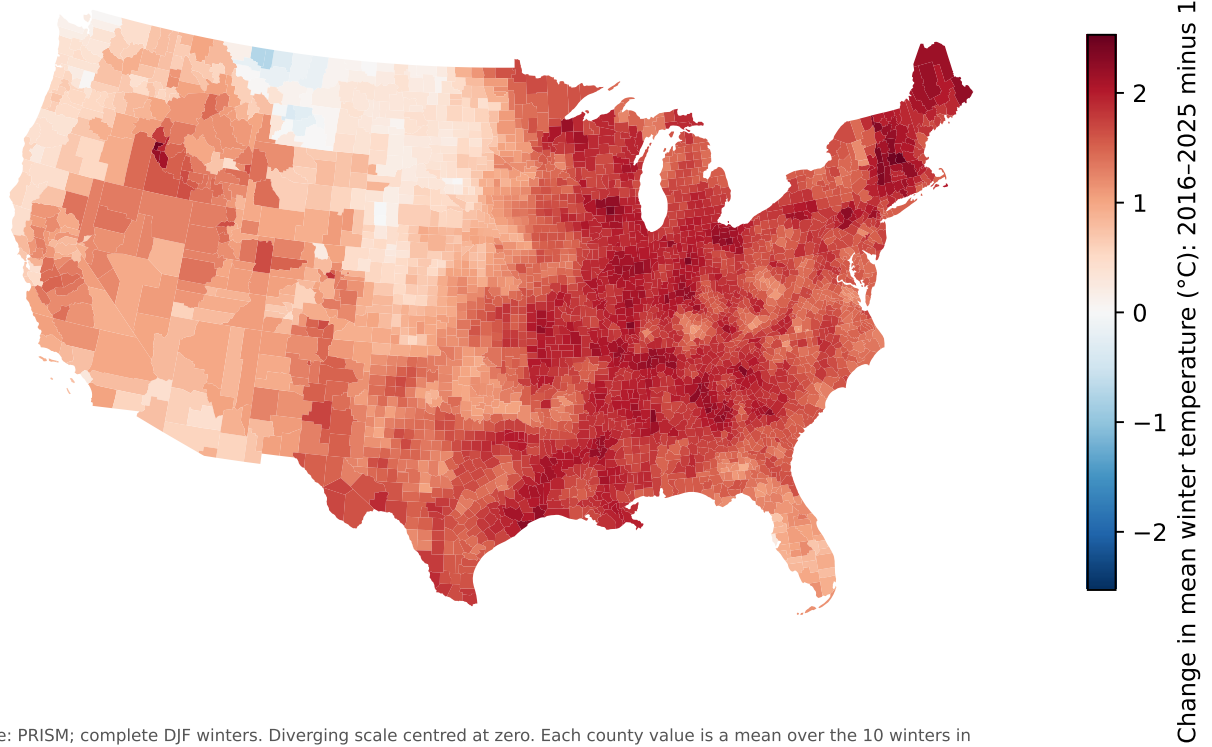
### 3 Where winters have warmed



Source: PRISM; complete DJF winters. Identical colour limits in both panels. Each county value is a mean over the 10 winters in its period (1982–1991 and 2016–2025), not a comparison of single winters or of sample endpoints.

**Figure 2:** Mean winter temperature in the early (1982–1991) and recent (2016–2025) periods, on identical colour limits. Both panels are ten-winter period means, not single-year endpoints.

### Where U.S. winters have warmed the most

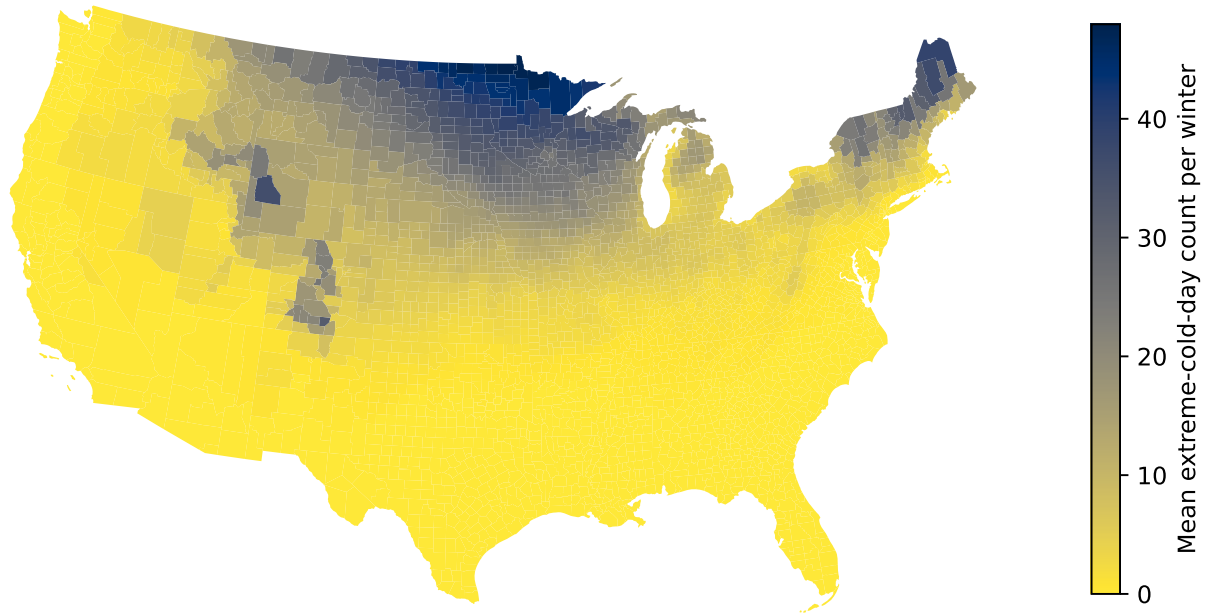


Source: PRISM; complete DJF winters. Diverging scale centred at zero. Each county value is a mean over the 10 winters in its period (1982-1991 and 2016-2025), not a comparison of single winters or of sample endpoints.

**Figure 3:** Change in mean winter temperature between the two periods. The diverging scale is centred at zero, so the map shows both the breadth of warming and its regional heterogeneity.

## 4 Cold extremes

Extreme-cold winter days are concentrated in the northern U.S.

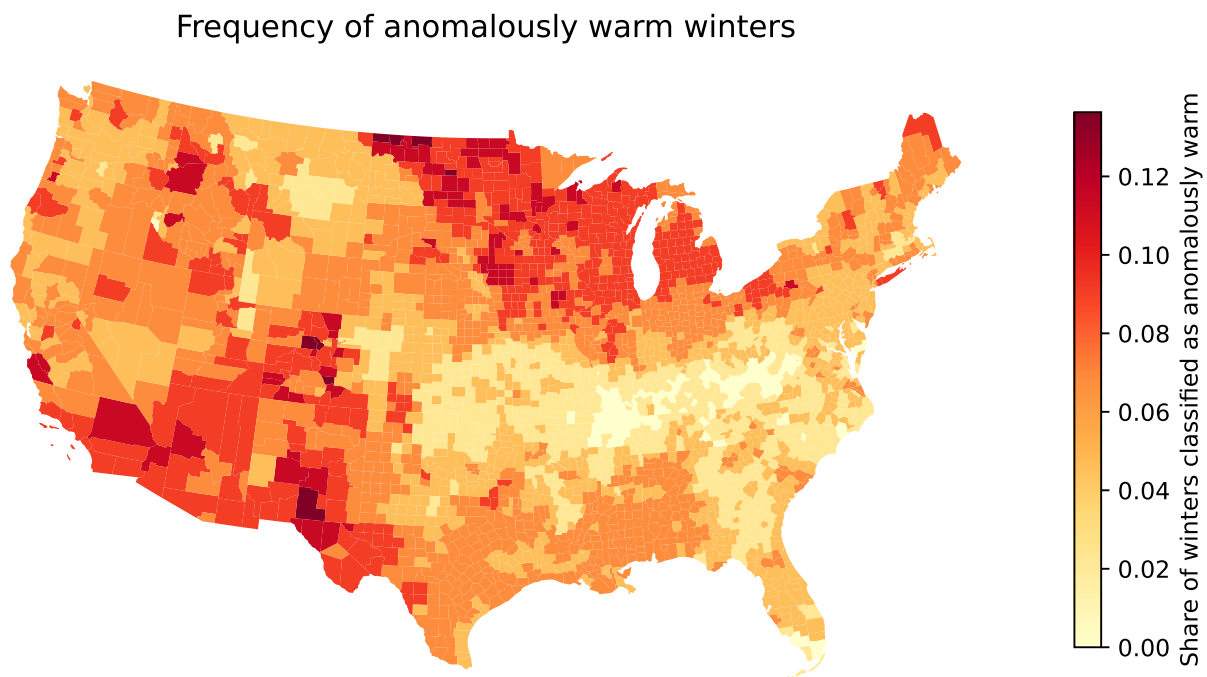


Source: PRISM; days with daily minimum temperature below 0°F, averaged over complete DJF winters. This is not a full Winter Severity Index.

**Figure 4:** Mean count of winter days with a daily minimum below 0°F. This is the extreme-cold measure only; it is not a Winter Severity Index.

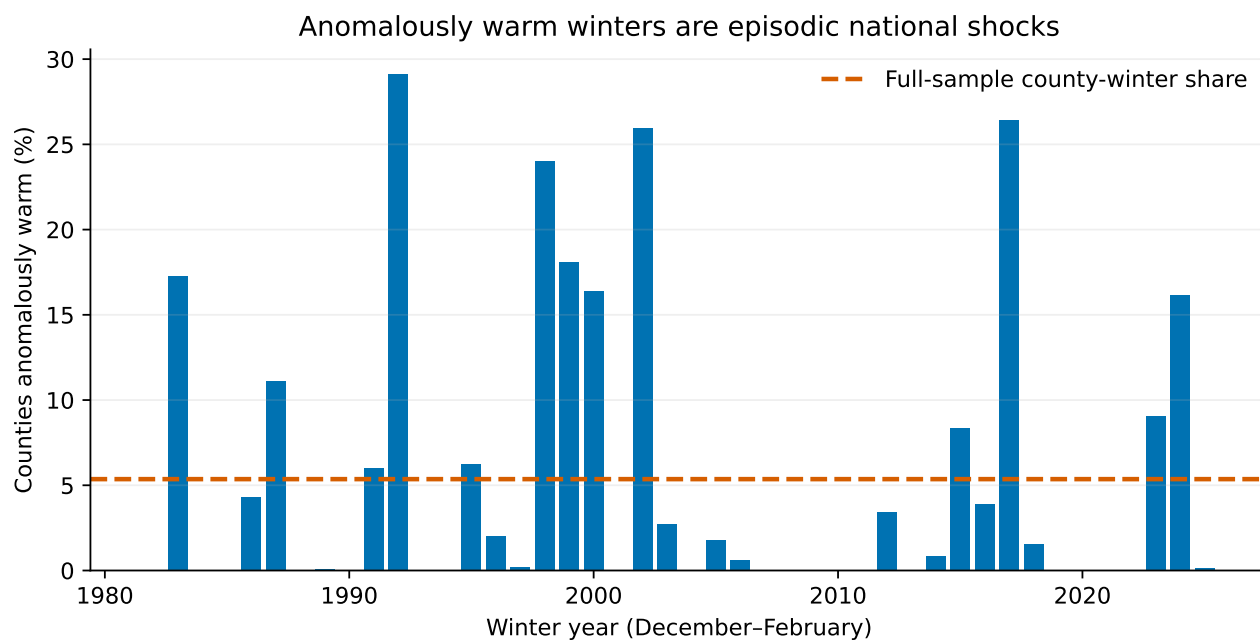


## 5 Anomalously warm winters



PROVISIONAL — requires Eyal's confirmation. County-specific detrended residual  $> 1.5$  SD; at least 30 complete winters. The definition uses the full sample to classify each winter and removes each county's trend, so mapped shares partly reflect residual tail shape. See the caveat table.

**Figure 5: Provisional.** Share of a county's eligible winters classified as anomalously warm, defined as a detrended residual above 1.5 standard deviations of that county's own full-sample distribution.



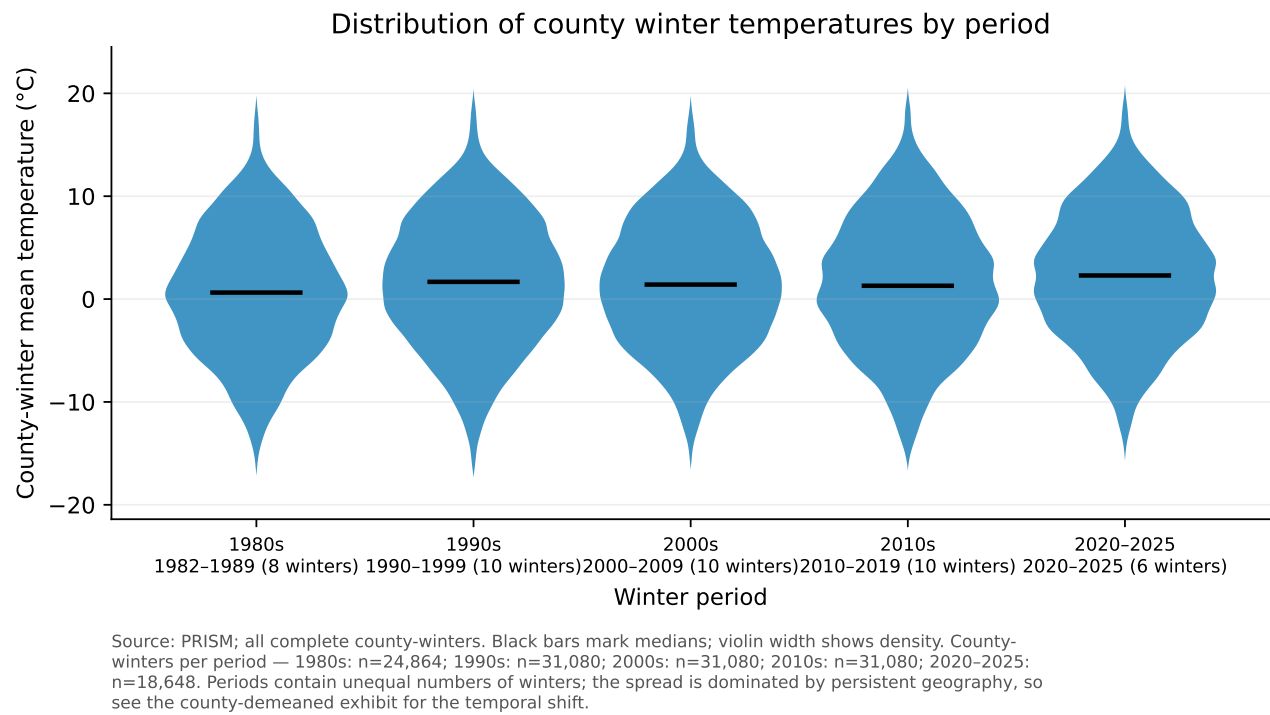
PROVISIONAL — requires Eyal's confirmation. Detrending separates short-run shocks from gradual county warming, and by construction removes the trend itself.

**Figure 6: Provisional.** Share of counties classified as anomalously warm in each winter. Because each county's linear trend is removed, this isolates episodic shocks from gradual warming.

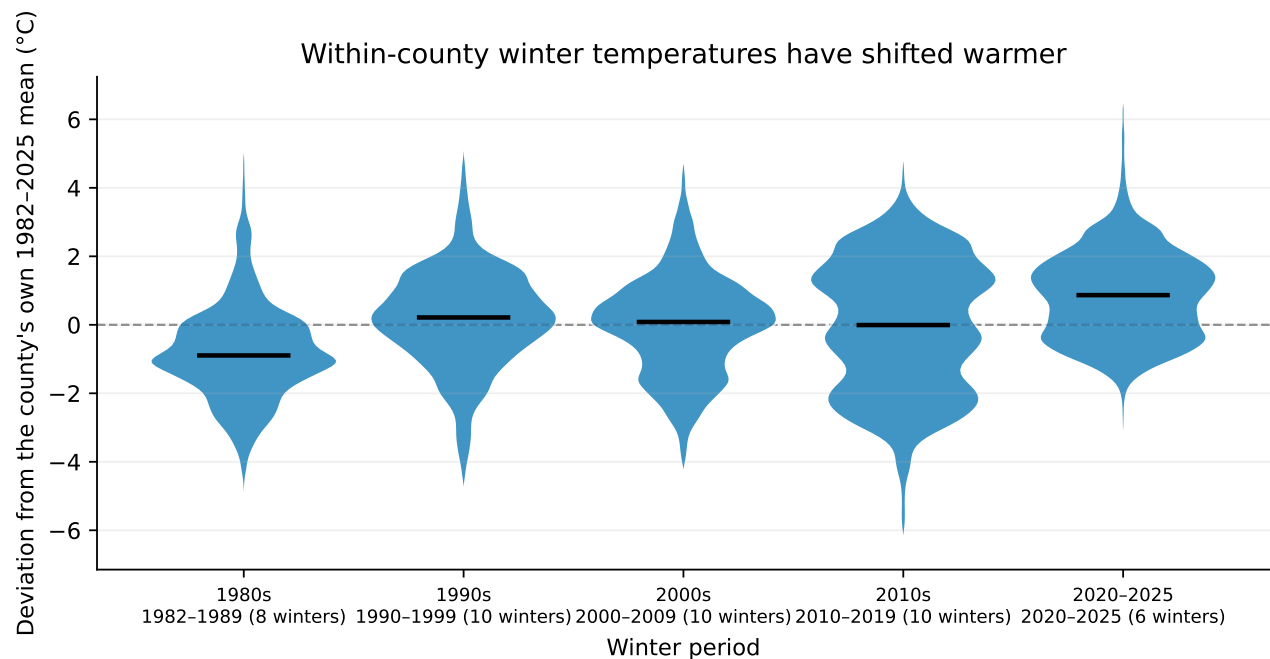
## **Known properties of the provisional definition**

1. Look-ahead: each county's reference distribution uses the full 1982–2025 sample, so later winters inform how earlier winters are classified.
2. Detrending removes each county's linear trend, so gradual warming is excluded from the shock measure by construction.
3. A county's mapped frequency partly reflects the skewness and tail shape of its residual distribution, not exposure to warm shocks alone.
4. The 1.5 SD cut-off is a convention, not an estimated threshold.
5. For regression use, compare against a fixed-baseline alternative (e.g. above the county's 1982–2000 mean by 1.5 baseline SD) or use the continuous standardised anomaly.

## 6 Distributional shift



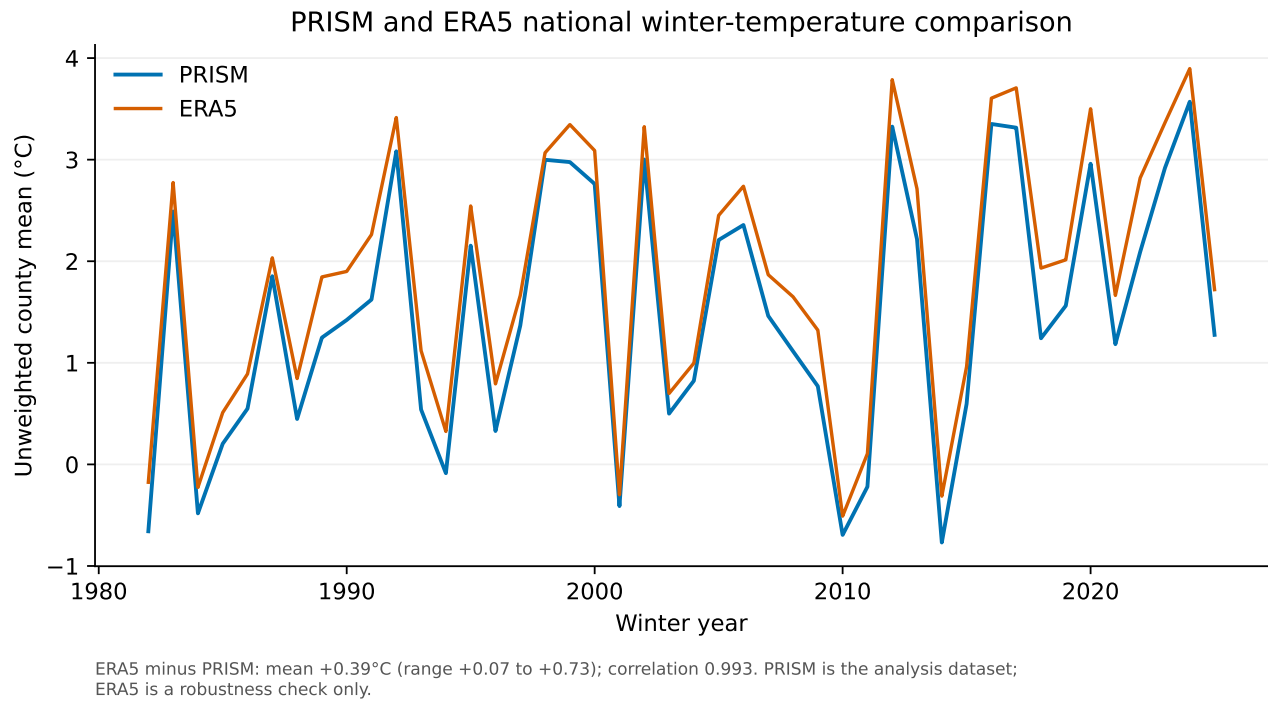
**Figure 7:** County-winter mean temperatures pooled by period. Periods contain unequal numbers of winters, and the pooled spread is dominated by persistent cross-county geography, which compresses the apparent temporal shift.



Source: PRISM; all complete county-winters, each expressed as a deviation from its own county's full-sample mean. Removing persistent geography isolates the temporal shift. County-winters per period — 1980s: n=24,864; 1990s: n=31,080; 2000s: n=31,080; 2010s: n=31,080; 2020-2025: n=18,648.

**Figure 8:** The same periods after subtracting each county's own full-sample mean. Removing persistent geography isolates the within-county shift, which is the quantity of interest.

## 7 Robustness: PRISM versus ERA5



**Figure 9:** National winter temperature from both products. PRISM is the analysis dataset; ERA5 is a targeted robustness check. The mean offset and correlation are reported on the figure.

## 8 Decisions requested

### Anomaly definition (regression use)

**Current choice.** Detrended county residual  $> 1.5$  SD of the full 1982–2025 sample; minimum 30 winters

**Issue.** Uses future winters to classify earlier ones and removes secular warming by construction; threshold is a convention. Alternatives: fixed 1982–2000 baseline, or the continuous standardised anomaly.

**Status.** CONFIRM — descriptive exhibits are labelled PROVISIONAL

### Geographic weighting

**Current choice.** Unweighted mean across counties

**Issue.** Estimand is the average county, not an area-, population- or road-weighted national average. A VMT- or road-mile-weighted series may be the relevant exposure weight for the collisions stage.

**Status.** CONFIRM

### Comparison periods

**Current choice.** 1982–1991 versus 2016–2025

**Issue.** Symmetric ten-winter period means; not 1981-versus-2025 endpoints.

**Status.** CONFIRM

### Extreme-cold exhibit

**Current choice.** Mean days with  $TMIN < 0^{\circ}F$ ; not labelled WSI

**Issue.** Not a full Winter Severity Index.

**Status.** DOCUMENTED

### ERA5 robustness exhibit

**Current choice.** Targeted national PRISM-ERA5 comparison

**Issue.** PRISM remains the analysis dataset; ERA5 runs warm by a small constant offset.

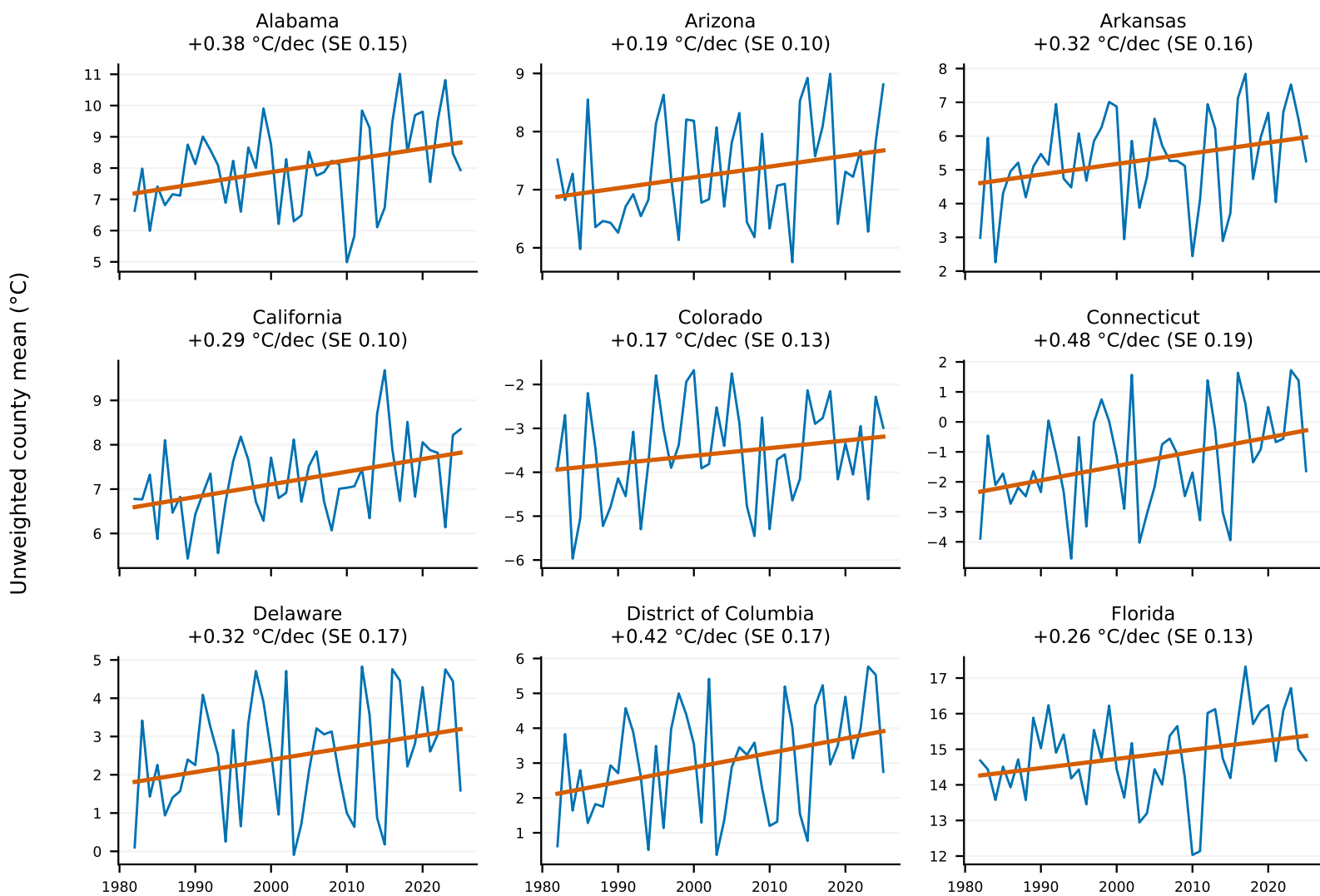
**Status.** DOCUMENTED

## A State-level winter-temperature trends

Per-state annual means with state-specific linear trends and the standard error of each slope. Reproduced in full; page count follows the run.

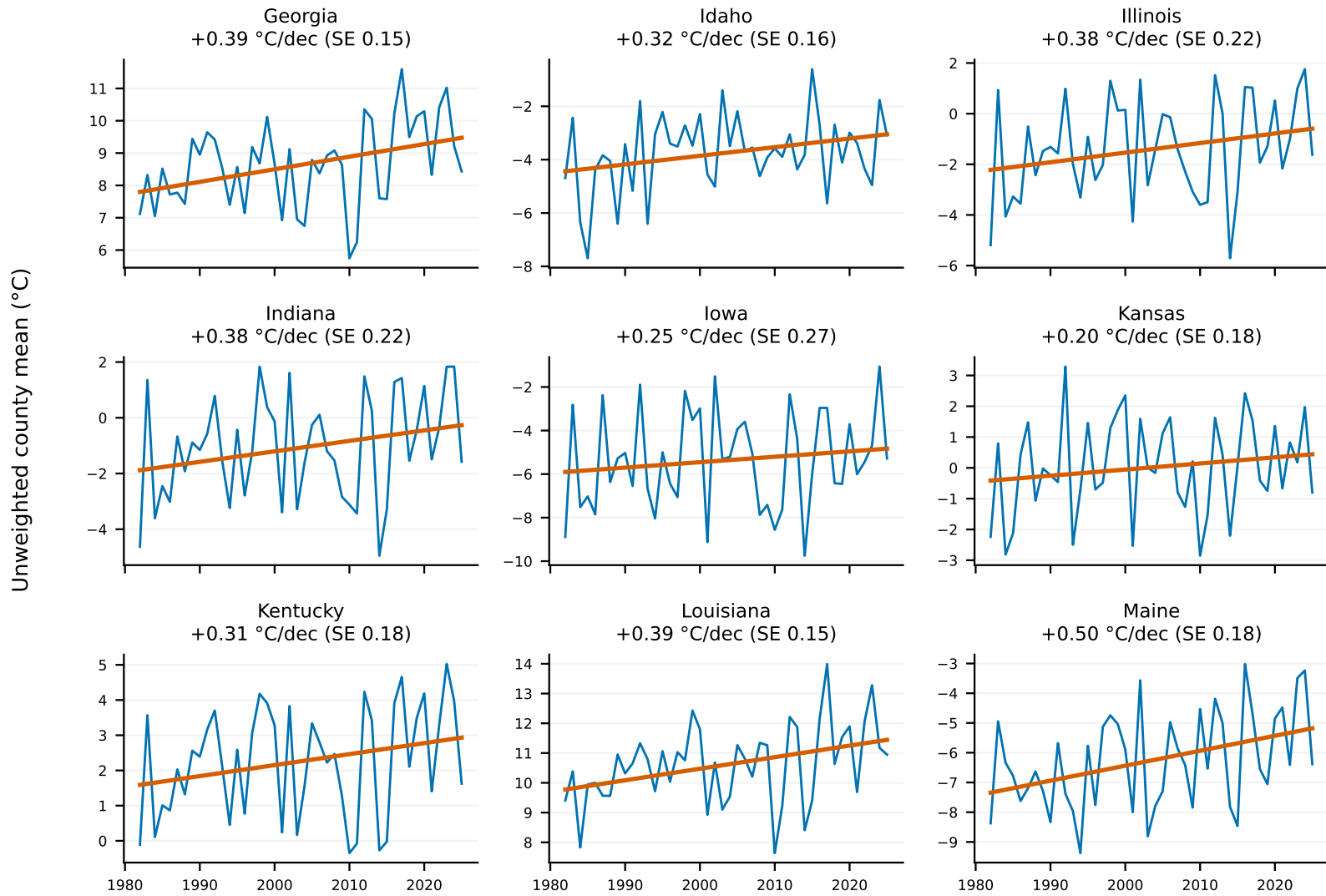


## Winter-temperature trends vary across states



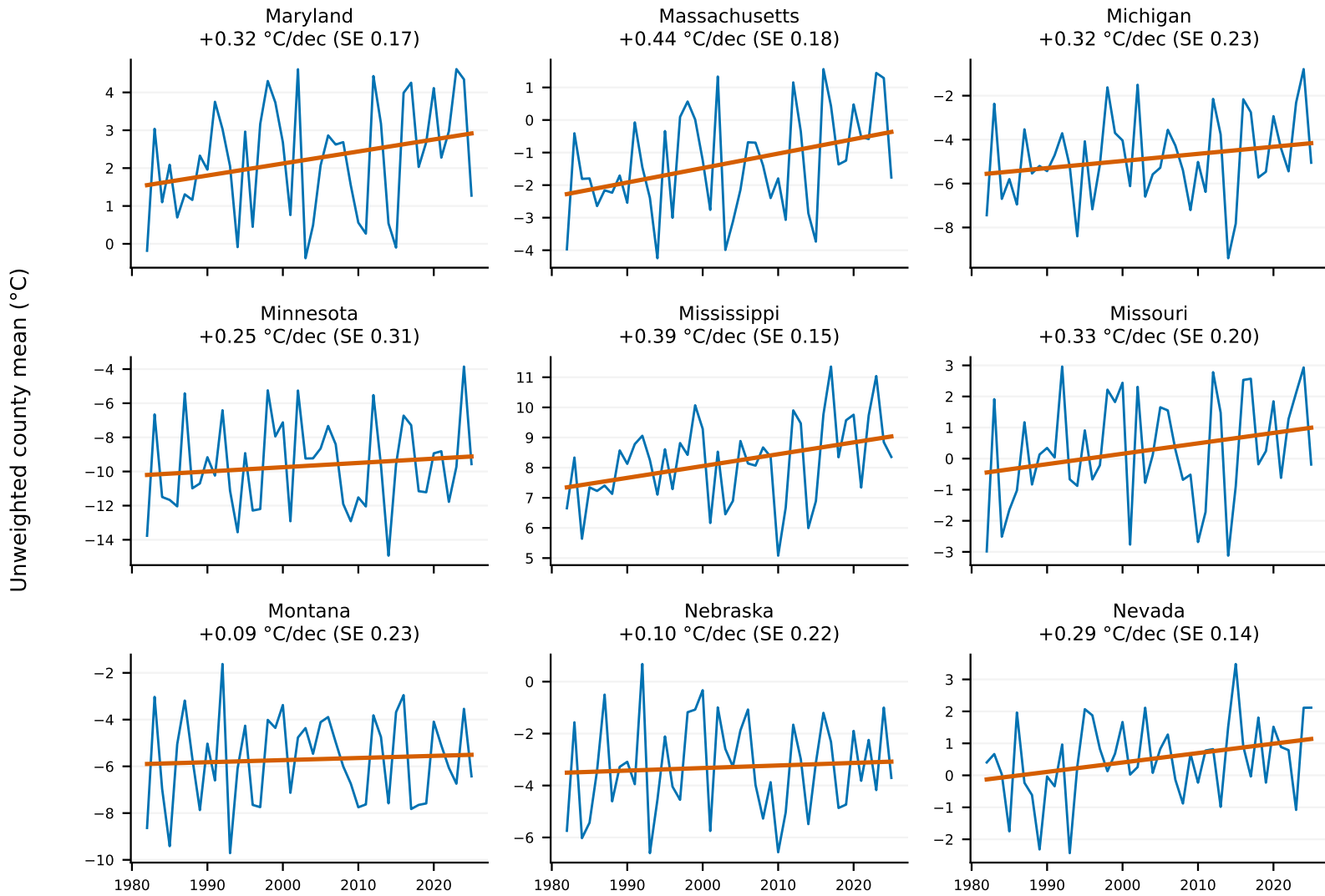
Source: PRISM; complete DJF winters. Orange lines are state-specific linear trends; SE is the standard error of the slope.

## Winter-temperature trends vary across states



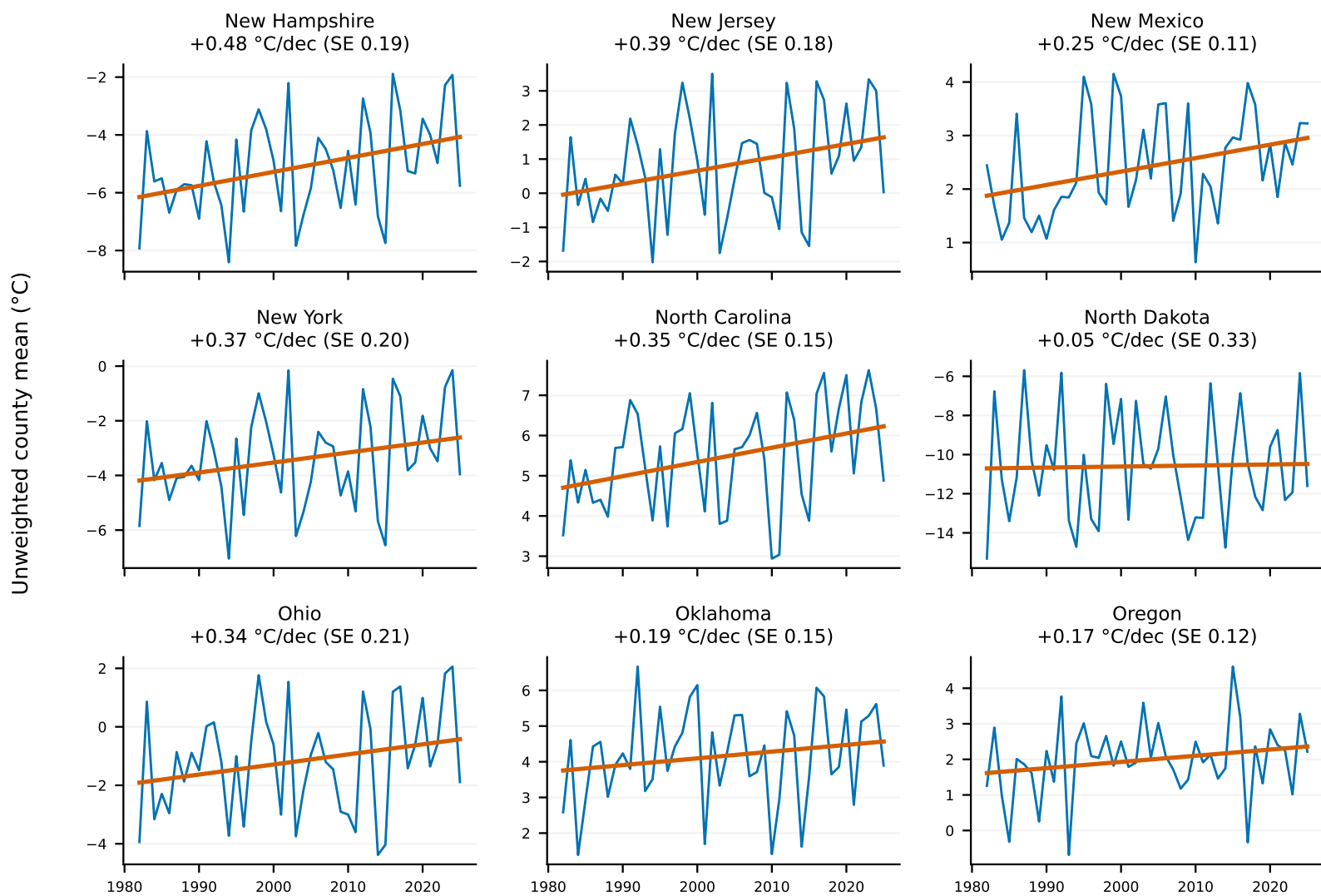
Source: PRISM; complete DJF winters. Orange lines are state-specific linear trends; SE is the standard error of the slope.

## Winter-temperature trends vary across states



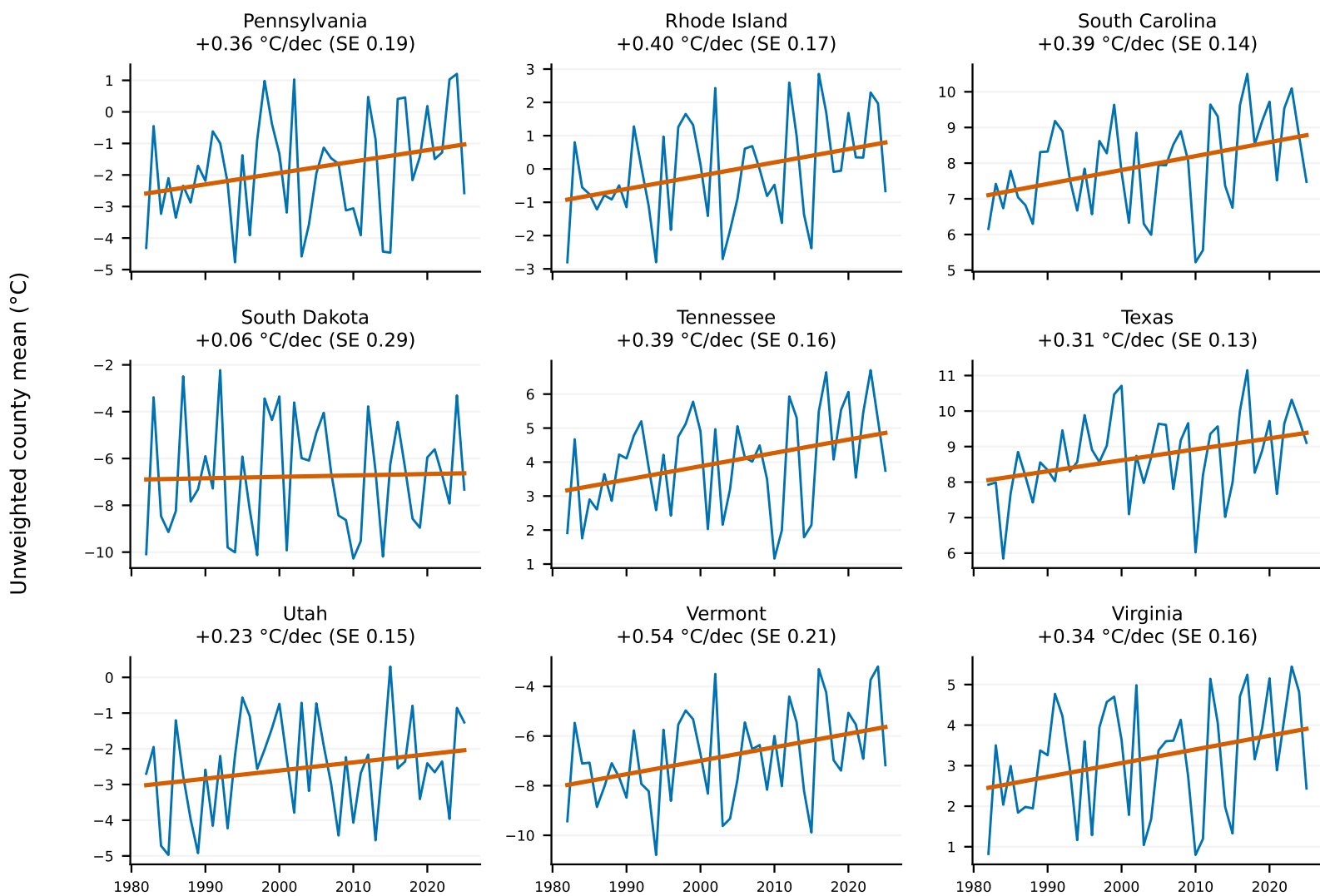
Source: PRISM; complete DJF winters. Orange lines are state-specific linear trends; SE is the standard error of the slope.

## Winter-temperature trends vary across states



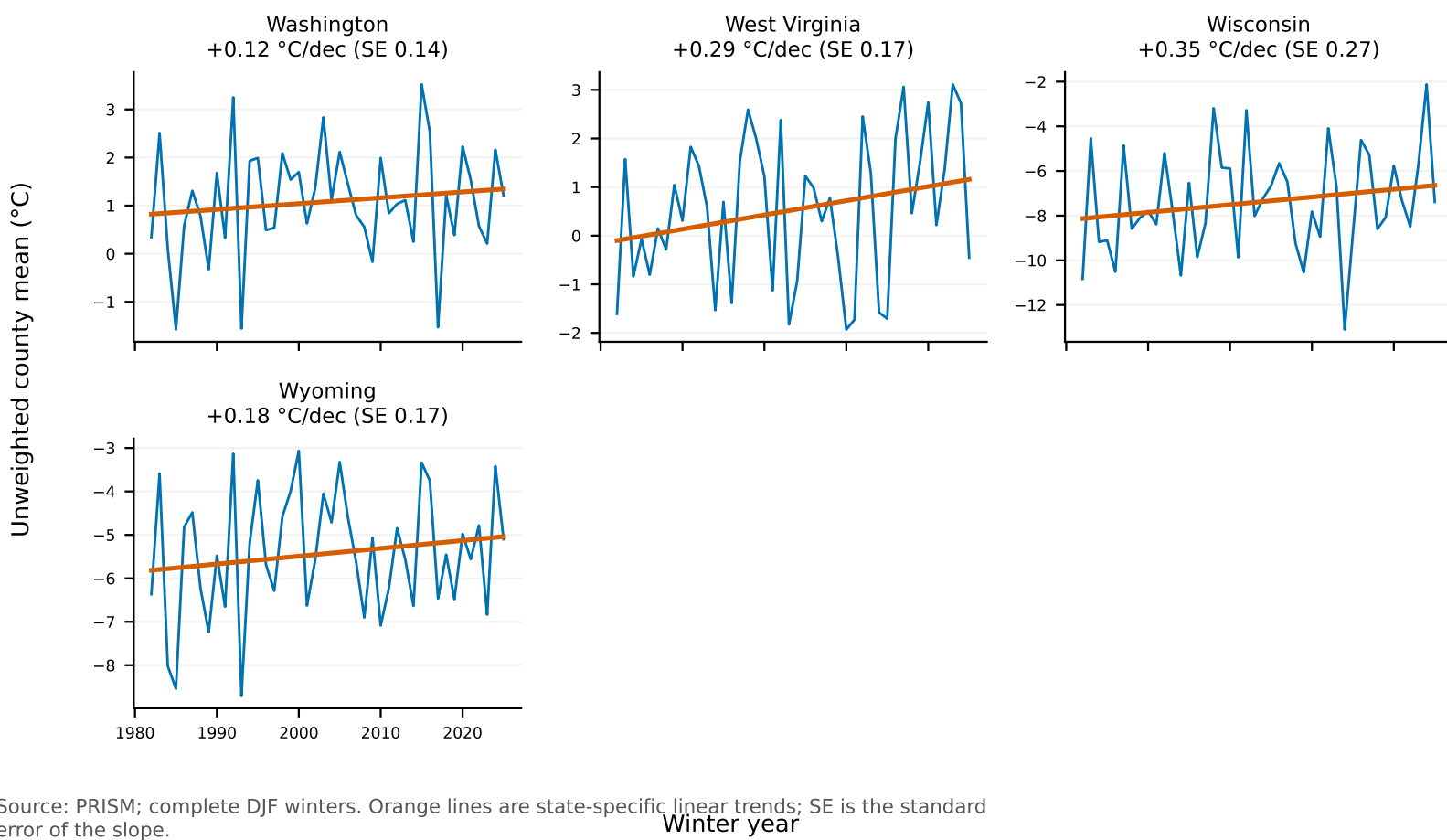
Source: PRISM; complete DJF winters. Orange lines are state-specific linear trends; SE is the standard error of the slope.

## Winter-temperature trends vary across states



Source: PRISM; complete DJF winters. Orange lines are state-specific linear trends; SE is the standard error of the slope.

Winter-temperature trends vary across states



## B Tier 1 QA artefacts (not reproduced)

The following internal QA exhibits and tables were produced by the same run but not included in this report.

- figures/weather/tier1/prism\_interannual\_variability\_choropleth.pdf
- figures/weather/tier1/prism\_national\_trend\_days\_below\_freezing\_32f.pdf
- figures/weather/tier1/prism\_national\_trend\_days\_extremely\_cold.pdf
- figures/weather/tier1/prism\_national\_trend\_freeze\_thaw\_days.pdf
- figures/weather/tier1/prism\_county\_spaghetti\_mean\_temp\_c.pdf
- figures/weather/tier1/prism\_state\_trends\_freeze\_thaw\_days.pdf
- figures/weather/tier1/prism\_state\_decade\_distributions\_mean\_temp\_c.pdf
- figures/weather/tier1/prism\_state\_trends\_mean\_temp\_c.pdf
- figures/weather/tier1/prism\_state\_trends\_days\_below\_freezing\_32f.pdf
- figures/weather/tier1/prism\_national\_trend\_mean\_temp\_c.pdf
- figures/weather/tier1/prism\_state\_trends\_days\_extremely\_cold.pdf
- tables/weather/tier1/prism\_tier1\_qa\_findings.csv
- tables/weather/tier1/prism\_summary\_by\_county\_review\_flags.csv
- tables/weather/tier1/prism\_summary\_pooled.csv
- tables/weather/tier1/prism\_interannual\_variability\_by\_state.csv
- tables/weather/tier1/prism\_summary\_by\_calendar\_month.csv
- tables/weather/tier1/prism\_tier1\_coverage\_integrity.csv
- tables/weather/tier1/prism\_summary\_by\_county.csv
- tables/weather/tier1/prism\_county\_interannual\_variability.csv